

Executive Summary: Public EV Charging Access in New Jersey

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Purpose

This project examines how public electric vehicle (EV) charging infrastructure is distributed across New Jersey census tracts. The main question is whether public charging access is evenly distributed or concentrated in particular places.

Data Used

The analysis combines public open EV charging station data, ACS census tract demographic data, and New Jersey census tract boundary shapefiles.

Key Facts

- The analysis includes 2,181 New Jersey census tracts.
- The filtered station data include 1,822 public open EV charging stations.
- These stations include 3,965 Level 2 ports and 2,062 DC fast ports, for a total of 6,027 charging ports.
- Only 701 tracts, or about 32.1%, contain at least one public EV charging station.
- Most tracts have zero public stations, while a small number contain several stations.
- Bergen and Middlesex Counties have the most public stations, followed by Essex, Hudson, and Morris.

Main Findings

- Public EV charging access is uneven across New Jersey census tracts.
- Charging access is modestly but statistically significantly spatially clustered.
- Tracts with stations have higher median household income than tracts without stations.
- Regression results show that median income is positively associated with station counts.
- Owner-occupancy is negatively associated with station counts.
- Poverty rate and no-vehicle household share are not strong predictors in the baseline model.
- Adding population density improves model fit, but spatial clustering remains in the residuals.

Interpretation

The results suggest that public EV charging infrastructure is concentrated rather than evenly distributed. Demographic variables, especially income, explain part of the pattern, but they do not explain everything. Remaining spatial patterns suggest that highways, commercial areas, parking availability, employment centers, tourism, utility capacity, and EV adoption may also matter.

The results show associations between tract characteristics and charging access, but they do not prove that income, housing patterns, or other demographic variables cause charger placement.

Summary

Public EV charging access in New Jersey is spatially uneven and clustered. About one-third of census tracts contain a public station. Income is associated with station availability, but geographic and land-use factors remain important for understanding where chargers are located.